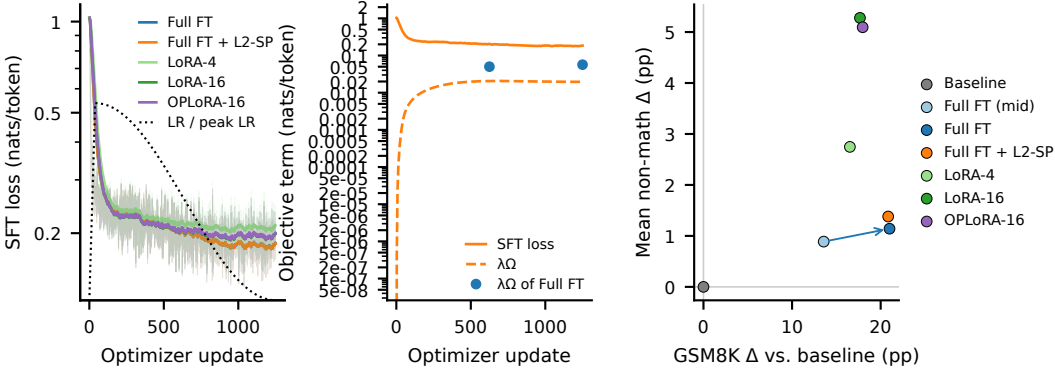

(a) SFT loss and LR schedule
SFT loss (nats/token)
1
0.5
0.2
0
500
1000
Optimizer update
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16
LR / peak LR
(b) L2-SP: penalty vs. loss
Objective term (nats/token)
2
1
0.5
0.2
0.1
0.05
0.02
0.01
0.005
0.002
0.001
0.0005
0.0002
0.0001
5e-05
2e-05
1e-05
5e-06
2e-06
1e-06
5e-07
2e-07
1e-07
5e-08
0
500
1000
Optimizer update
SFT loss
λΩ
λΩ of Full FT
(c) Adaptation vs. retention
Mean non-math Δ (pp)
5
4
3
2
1
0
0
10
20
GSM8K Δ vs. baseline (pp)
Baseline
Full FT (mid)
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16